PROJECT 2 — SCHOOL IT EQUIPMENT INVENTORY ANALYSIS

Data cleaning, spreadsheet calculations, charts and SQL companion

Source: Cortizo_Case Study8 organize.xlsx, supplied by student **Analyzed sheet:** Cleaned Data, 50 valid asset records (A8000–A8049) **Supporting deliverables:** Reviewed Excel copy, CSV extract, SQLite database, SQL script, original workbook (unchanged).

1. Problem and goals

Schools need to know the number, condition and estimated cost of IT equipment by type and department. This case study summarizes the provided classroom dataset to identify inventory concentrations and possible repair priorities.

2. Data schema and tools

Ten fields: Asset_ID, Equipment_Type, Brand, Condition, Age_Years, Quantity, Unit_Cost, Maintenance_Cost, Warranty, Department. Excel/Google Sheets-style formulas, Excel pivot tables and charts were used; an additional SQL companion database now demonstrates actual SQL queries against the same cleaned records.

3. Data preparation

The source workbook includes Raw_Data, Cleaned Data, Calculations, Pivot Table, Chart, and Interpretation and recommendati sheets. Reviewing the complete cleaned records found **50 records**, **50 unique IDs**, no blank values in the main ten data fields, and **no duplicate Asset_IDs**. The source worksheet's old duplicate-check cell contained 1, contradicting the actual records; this is corrected in the **separate reviewed workbook copy**.

4. Calculated results from all 50 cleaned records

Measure	Result
Asset records	50
Distinct asset IDs	50
Sum of equipment quantities	279
Estimated acquisition value (Quantity * Unit_Cost)	PHP 9,168,000
Recorded maintenance costs (sum of row entries)	PHP 115,182
Records marked Needs Repair	12
Units within Needs Repair records	74
Fair-condition records aged 5+ years	8

Equipment unit counts: Router 70; Projector 69; Laptop 60; Desktop 49; Printer 31. **Department unit counts:** Admin 98; IT 74; Faculty 61; Library 46. **Condition unit counts:** Good 108; Fair 97; Needs Repair 74. **Maintenance by type (sum of row entries):** Laptop PHP 28,361; Projector PHP 23,488; Router PHP 22,238; Desktop PHP 21,890; Printer PHP 19,205.

5. Analysis approach

- Count or sum equipment quantity by category and department using SUMIF, SUMIFS and pivot tables.
- Use SUMPRODUCT(Quantity, Unit_Cost) to calculate an estimated total acquisition value; summing unit costs alone is not the same thing.

- Use mean, median, minimum and maximum to summarize age, quantity and costs.
- Compare repair quantities and maintenance totals to propose possible priorities.
- Use the chart worksheet to present equipment distribution visually.

6. Important quality corrections

The **original workbook is preserved** for transparency. It contained partial or inconsistent results (e.g., Calculations!B9 showed 270 units, B10 showed PHP 8,763,000, Router quantity was listed as 61 in one place, and Cleaned Data!B54 suggested one duplicate despite 50 unique Asset_IDs). The revised workbook recalculates the relevant summary cells from the full 50 cleaned records and retains workbook charts and the pivot table.

The original narrative interpretation also made claims that do not match the underlying dataset, such as **zero** laptops in IT (the data shows seven units). Treat those original narrative statements as drafts. Use the current facts and recommendations below in your presentation.

7. Recommendations grounded in the data

- Prioritize inspecting the 12 Needs Repair asset records covering 74 units; evaluate condition and replacement costs before purchasing.
- Assess printers by department. The dataset shows 20 printer units assigned to IT and none in Library; confirm actual usage before moving equipment.
- Review aging fair-condition equipment and plan phased maintenance or replacements, since eight fair-condition records have age at least five years.
- Review the categories with higher recorded maintenance totals, especially laptops, and compare repeated costs against replacement options.

These are analytical recommendations, **not evidence that a school actually reallocated equipment**.

8. SQL/database evidence

assets/Equipment_Analysis.sqlite and documentation/Equipment_Analysis_Database.sql contain the 50-row relational table and queries. Examples:

```
SELECT SUM(quantity) AS total_units,
    SUM(quantity * unit_cost) AS estimated_acquisition_value,
    SUM(maintenance_cost) AS maintenance_cost
FROM equipment;

SELECT equipment_type, SUM(quantity) AS total_units
FROM equipment GROUP BY equipment_type
ORDER BY total_units DESC;

SELECT COUNT(*) AS records, SUM(quantity) AS affected_units
FROM equipment WHERE condition = 'Needs Repair';
```

Outputs: 279 units, PHP 9,168,000 acquisition, PHP 115,182 maintenance; 12 repair records covering 74 units. SQL is portable to SQLite and can be demonstrated in DB Browser for SQLite or SQLite's CLI.

9. Testing and limits

Checked totals independently from all 50 cleaned rows and against the workbook's original Pivot Table summary. Formula-driven cells were reimported for verification. It is a classroom dataset: no actual school inventory access, asset purchase authorization or real repair activity was verified. Maintenance cost is recorded per source row and should not be assumed to be the per-unit cost without additional definition.